Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	Asian Sentences Is Amazingly Not Sane (ASIANS)
Project / Solution Name	InstaQuote
Name (Leader)	Teng Min Jie
Email & Phone (Leader)	tengminjie268@gmail.com (+60182658208)
University Affiliation(s)	Asia Pacific University of Technology & Innovation
Live Prototype URL	https://instaquote-orbyte.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/PikaPizza495/SDC_Hackathon_Submission_Asian_Sentences_Is_Amazingly_Not_Sane

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Our system connects end-to-end as a cloud-hosted web application on Replit. First, the user interacts with our Streamlit frontend, which is styled with custom CSS to match the Orbyte dashboard interface. When a low-stock alert is triggered from our simulated Orbyte hub (orbyte_hub.py), the Python backend uses Pandas to read our supplier data from PostgreSQL (with an automatic fallback to our local suppliers.csv file). The backend then runs an explainable scoring algorithm to rank the best Malaysian vendors based on budget, minimum order quantity (MOQ), lead time, and eco-certification. Next, when drafting a quotation request, the backend sends a short JSON request via Python's built-in urllib library to an external LLM API (Gemini, Groq, or OpenRouter). Finally, after the SME Owner approves the draft through our Role-Based Access Control (RBAC) step, the app logs the event and generates a pre-filled WhatsApp wa.me link.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11+	Core programming language for backend logic, scoring algorithm, and REST API calls (urllib).	PSF License
Streamlit 1.32.0	Web application framework used to build the interactive Orbyte procurement UI and manage session state.	Apache License 2.0
Pandas 2.0.0	Data manipulation library used to load, filter, and score the Malaysian supplier dataset.	BSD 3-Clause License
psycopg2-binary 2.9	PostgreSQL database adapter used to query suppliers and store multi-tenant activity audit logs.	LGPL v3 License

Name & Version	Purpose in Project	License
Google Fonts	Typography imported via CSS to match the native Orbyte user interface design.	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API	Generates bilingual (English and Bahasa Melayu) WhatsApp Request for Quotation (RFQ) messages.	Free
Groq Cloud API	Alternative high-speed LLM provider for generating WhatsApp RFQ drafts.	Free
OpenRouter API	Alternative routing API locked strictly to free tier models for generating RFQ drafts.	Free
WhatsApp Click-to-Chat (wa.me / Meta)	Generates pre-filled WhatsApp message deep links to dispatch approved RFQs to suppliers.	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit (Replit Deployments)	Cloud IDE used for building, previewing, and hosting the live Streamlit web application.	Sponsor-Provided (Replit Core)
Replit PostgreSQL Database	Cloud relational database for storing supplier records and audit logs (with CSV/session fallback).	Sponsor-Provided (Replit Core)
GitHub	Version control, source code management, and public code repository hosting for submission.	Free

Service & Provider	Purpose in Project	Plan / Region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-2.0-flash	Google Gemini API	Pre-trained (called via REST API)	Google APIs Terms of Service
llama-3.1-8b-instant / llama-3.3-70b-versatile	Groq Cloud API	Pre-trained (called via REST API)	Llama 3.1 / 3.3 Community License
openai/gpt-oss-20b	Groq Cloud API	Pre-trained (called via REST API)	Apache License 2.0
meta-llama/llama-3.3-70b-instruct:free	OpenRouter API	Pre-trained (called via REST API)	Llama 3.3 Community License
google/gemma-3-27b-it:free	OpenRouter API	Pre-trained (called via REST API)	Gemma Terms of Use

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Malaysian SME Suppliers Dataset (suppliers.csv)	Generated via our team's generate_suppliers.py script in the GitHub repo	Synthetic	Apache License 2.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Google Gemini	Writing Python application code, building the supplier scoring logic, and styling the custom CSS interface.	Major
Anthropic Claude	Debugging code issues, fixing errors, and expanding product feature ideas for the procurement workflow.	Moderate
DeepSeek	Learning how to use GitHub, Git version control workflows, and repository setup.	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Our team defined all product features, procurement workflows, and UI requirements ourselves before prompting any AI tools. Whenever Gemini or Claude generated Python code or CSS styling, we manually tested the output inside VS Code and Replit to check for bugs, UI glitches, and broken API calls. When we spotted issues such as LLM provider connection errors or layout problems, we actively debugged the behavior, verified that our scoring logic and RBAC rules worked accurately, and confirmed full compliance before committing to GitHub.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Replit Core Platform & Cloud Hosting	Used to preview, test, and publish the live Streamlit web application and manage database/environment configurations.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was very helpful for cloud previewing, Git syncing, and one-click live deployment so judges can test our application online. However, it is not strictly locked to our core functionality and could be replaced if needed. Because our codebase is built in standard Python and Streamlit with an automatic local CSV fallback (suppliers.csv), the exact same application can run locally or on any standard Python hosting platform (such as VS Code).

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No user data is collected. Our application uses 100% synthetic supplier data (suppliers.csv) where all company names, phone numbers, and @example.com email addresses are fictional. When a user or judge enters an optional LLM API key in the sidebar to test AI message generation, the key is held only in temporary browser memory for that single HTTP request, has browser autocomplete turned off, and is immediately cleared after use. API keys are never logged, never saved to Replit or PostgreSQL, and never committed to our GitHub repository.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The main technical risks of using external LLM APIs are rate limits on free plans, provider downtime, accidental token drain, and AI hallucinations (such as inventing wrong prices or leaving blank placeholders in the message). To mitigate these risks, we added a strict output limit of 180 tokens per request, session caching so repeated clicks use zero tokens, and a hard cap of 50 AI calls per session. We also locked OpenRouter strictly to free-tier models ($0.00/token) to prevent accidental billing. Most importantly, if no API key is entered or an API fails, the app automatically falls back to a built-in bilingual template.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]
24/10/2026